

Term paper instructions for class: Introduction to Text Mining and Natural Language Processing

Goal of the term paper is to convince me that you have understood the material that we discussed in class and are able to integrate with a topic that you are interested in. You will form teams of up to 3 students to complete the task. You should write between 5 and 10 pages describing main idea of the project, the motivation for your question, the text data you used and results.

I function through Figures/Tables. Good figures are key. Don't make figures that are not necessary and you cannot explain. This is a general point: being able to "explain" your own results is key – don't show things you don't reason about. Don't flood me with contradictory evidence without explaining.

Start slow end high. It is best practice to start with a description of the data, summary statistics, some figure giving a meaningful summary of elements of the data you care about/that fits to what you want to show.

Term-paper preliminary results presentation March 19th. Term paper deadline on March 31st. In your presentation you are supposed to discuss the main idea of the project, the motivation for your question, the text data you want to use (and results). I will ask questions and try to give feedback. I will keep track of the feedback I give and will watch out how you reacted to it in the thesis.

Hand-in of the pdf containing the term paper, the codes used and the raw data. Send one zip file with everything in. If the data is large do not send but you should send jupyter notebooks that allow me to follow the code output if possible.

Additional points:

I know you have little time. Hand-in is in a month from now but you have several term papers. So effectively you have like 5 days to work on this. But people have done AMAZING thing in this course so my bar is high (sorry).

Creativity wins. Human creativity will be rewarded – especially because it is hard with so little time.

Optimally you use one of the methods we discussed in class to generate the text (i.e. scraping, reading files, pre-processing, tf-idf, dictionaries, LDA, supervised learning). Do NOT use LLMs for data generation or analysis.

Optimally you pay special attention to pre-processing steps and discuss these in your code and paper and do some analysis that shows what changes when you change pre-processing steps.

Optimally you do not only show something descriptive but derive some testable hypothesis that you test on the data you generate, or show the performance of a classifier you trained on the text data. I dislike clueless descriptive work even if technically done well.

This brings me to the most important point: integration. You should have a good idea by now what I mean by this term. Topic motivation, literature review, theory, data presentation, analysis, conclusion – there should be a red line going through these elements. This is integration and one dimension of grading will be whether you manage this.